Finite-n crossover in reduced-rank regression

B9: proof of Conjecture 1 for reduced-rank regression

Francis Headley

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Notation follows `outline.tex` and `derivations.tex`. Each statement carries one of the labels proved, cited, checked numerically (script in `code/`), or heuristic. All checks are in `code/b9_checks.py`; its output is in `code/out/b9_h1.txt` and `code/out/b9_h2.txt`.

1 Setting and results

The model is $y = ABx + \varepsilon$ with $A \in \mathbb{R}^{N\times H}$, $B \in \mathbb{R}^{H\times M}$, standard Gaussian priors φ, and

$$Z_n(C) = \int e^{-\frac{n}{2}\|AB-C\|_F^2}\varphi(A)\varphi(B)\,dA\,dB, \qquad R_n(C) = \frac{Z_n(C)}{Z_n(0)}, \qquad S = \tfrac{n}{2}CC^\top. \tag{1}$$

Let $q = \min(N,H)$, $p = \max(N,H)$, $a = \frac{1}{2}(p-q-1) \geq -\frac{1}{2}$, and for $0 \leq j \leq q$

$$\lambda_j = \tfrac{1}{2}\big[jM + (H-j)(N-j)\big], \qquad \kappa_j = \lambda_{j+1} - \lambda_j = \tfrac{1}{2}\big(M-N-H+2j+1\big) \quad (0 \leq j < q). \tag{2}$$

Since $\kappa_{j+1} = \kappa_j + 1$, λ_j is strictly convex and its minimum is attained at one index j_* ($m = 1$) or at two adjacent indices j_0, j_0+1 ($m = 2$, $\kappa_{j_0} = 0$) (proved in `derivations.tex`, Sec. 3.1). For $0 \leq j \leq q$ define the stratum profile

$$\Phi_j(S) = \mathbb{E}_U\,\mathbb{E}_V \exp\Big[-\tfrac{n}{2}\big\|(I-P_U)\,C\,(I-P_V)\big\|_F^2\Big], \tag{3}$$

with U Haar on $V_j(\mathbb{R}^N)$ and V Haar on $V_{H-j}(\mathbb{R}^M)$, independent, and $P_V := I_M$ when $H-j \geq M$. Φ_j depends on C only through the eigenvalues of S. For rank-one $C = \delta uv^\top$, $\|u\| = \|v\| = 1$, we write $s = n\delta^2/2$ and $\Phi_j(s)$.

The results are as follows.

1. Theorem 5 (proved, $m = 1$, every rank of C, all N, H, M): $|\log R_n(C) - \log \Phi_{j_*}(S)| \leq C_0\big(\|C\|_F^{2\gamma} + n^{-\gamma/2}\big)$ for $\|C\|_F \leq c_0$, $n \geq n_0$, with $\gamma > 0$ given in Eq. (13). The upper bound $\log R_n \leq \log \Phi_{j_*} + C_0 n^{-\gamma}$ holds for every C.

2. Remark 6 (proved for $(N,H,M) = (2,1,1)$, checked numerically for two other cases): at fixed $\delta > 0$ the error does not vanish as $n \to \infty$. For $(2,1,1)$ it tends to $-\delta$.

3. Corollary 8 (proved): for rank-one C, $\log \Phi_{j_*}(s) = -[\lambda(1) - \lambda(0)]\log s + [m(1)-1]\log\log s + O(1)$, with $\lambda(1)$ and $m(1)$ obtained independently of Ref. [1] and equal to the values of `dln.lam` in all 8000 cases $N, H, M \leq 20$ (checked numerically).

4. Theorem 9 (proved, $m = 2$, rank-one C): $R_n = \Phi_{j_0+1}(s)\,[\log \frac{n}{1+s} + O(1)]/[\log n + O(1)]$ uniformly in $0 \leq s \leq n/2$.

1.1 Corrections to Conjecture 1

Conjecture 1 states $\log R_n = \log \Phi_{j_*}(s) + o(1)$ as $n \to \infty$ uniformly in $\delta \geq 0$. The results above change it in four places.

1. The error is $o(1)$ in the joint limit $n \to \infty$, $\delta \to 0$, with the rate of Theorem 5. At fixed δ it converges to a nonzero function of δ (Remark 6). This is consistent with outline Sec. 1, where $c_\beta = (\lambda_1 - \lambda_0) \log \delta^2 + O(1)$; the $O(1)$ depends on δ.

2. For $m = 2$ the profile carries the factor $\log(n/(1+s))/\log n$. The error is $O(1/\log n)$ only for $s = n^{o(1)}$. At $s = n^\theta$, $0 < \theta < 1$, $\log R_n - \log \Phi \to \log(1-\theta)$, and at fixed δ the difference diverges as $-\log\log n$. This is the change of the power of $\log n$ between $S = 0$ and fixed C left open in `results_phase1_2.md`, Sec. 6.

3. The large-s form of $\log \Phi_{j_*}$ contains $+\log\log s$ when the rank-one multiplicity is $m(1) = 2$.

4. The restriction $H - j_* \leq M$ is not needed. If $j_* < q$ then $M - H + j_* \geq N - j_* \geq 1$; if $j_* = q = N$ then $\Phi_{j_*} \equiv 1$ (Lemma 3).

2 Exact representation

Lemma 1 (proved)**.** *Let $W_r = \{y \in \mathbb{R}^r : y_1 \geq \cdots \geq y_r \geq 0\}$, and for $y \in W_q$ let*

$$G_n(y) = \prod_{i=1}^{q} y_i^a (1+y_i)^{-M/2} e^{-y_i/(2n)} \prod_{i<l} (y_i - y_l), \qquad E(y) = \operatorname{diag}\big((1+y_1)^{-1}, \dots, (1+y_q)^{-1}, 1, \dots, 1\big) \in \mathbb{R}^{N\times N}, \tag{4}$$

and $h_S(y) = \mathbb{E}_O \exp[-\operatorname{tr}(S\,O\,E(y)\,O^\top)]$ with O Haar on $O(N)$. Then

$$R_n(C) = \frac{\int_{W_q} G_n(y)\, h_S(y)\, dy}{\int_{W_q} G_n(y)\, dy}. \tag{5}$$

Proof. The B-integral factorises over columns. With $Q = I_H + nA^\top A$ and the push-through identity $I_N - nAQ^{-1}A^\top = (I_N + nAA^\top)^{-1}$ it gives $\det(I_H + nA^\top A)^{-M/2} \exp[-\operatorname{tr} S(I_N + nAA^\top)^{-1}]$ (`derivations.tex`, Sec. 1.1). For Gaussian A, $AA^\top = O\operatorname{diag}(\ell_1, \dots, \ell_q, 0, \dots, 0)O^\top$ with O Haar on $O(N)$, independent of $\ell \in W_q$, and ℓ has density proportional to $\prod_i \ell_i^a e^{-\ell_i/2} \prod_{i<l}(\ell_i - \ell_l)$ (cited: Ref. [2], Thms. 3.2.17–3.2.18). Then $\det(I_H + nA^\top A) = \prod_i (1 + n\ell_i)$ and $(I_N + nAA^\top)^{-1} = O\,E(n\ell)\,O^\top$. The substitution $y = n\ell$ multiplies the density by a constant depending on n only, which cancels in the ratio. □

We use two facts. (M) For $S \succeq 0$ and diagonal $0 \leq D \leq D'$, $\operatorname{tr}(SODO^\top) \leq \operatorname{tr}(SOD'O^\top)$, and $\operatorname{tr}(SODO^\top) \leq \|D\|_\infty \operatorname{tr} S$. (W) If $f(\alpha)$, $\alpha \in \mathbb{R}^{p'\times r'}$, depends only on the nonzero eigenvalues $y \in W_r$ of $\alpha^\top\alpha$, $r = \min(p', r')$, then $\int f\, d\alpha = c \int_{W_r} f \prod_l y_l^{(|p'-r'|-1)/2} \prod_{l<l'}(y_l - y_{l'})\, dy$, and under any probability density of this form the left singular frame of α is Haar and independent of y (cited: Ref. [2], Sec. 3.2, by reweighting the Gaussian case).

3 Stratum integrals

For $0 \leq j \leq q$ and $r = q - j$ define on W_r

$$g^{(j)}(y) = \prod_{l=1}^{r} y_l^a (1+y_l)^{-M/2} \prod_{l<l'} (y_l - y_{l'}), \qquad h_S^{(j)}(y) = \mathbb{E}_O \exp\big[-\operatorname{tr}\big(S\,O\,E^{(j)}(y)\,O^\top\big)\big], \tag{6}$$

with $E^{(j)}(y) = \mathrm{diag}(0_j, (1+y_1)^{-1}, \dots, (1+y_r)^{-1}, 1_{N-q})$, and $Z^{(j)} = \int_{W_r} g^{(j)}$; for $j = q$, W_0 is a point and $g^{(q)} = Z^{(q)} = 1$. For $1 \le j \le q$ define on W_j

$$F^{(j)}(z) = \prod_{i=1}^{j} z_i^{a-M/2+q-j} e^{-z_i/2} \prod_{i<i'} (z_i - z_{i'}), \qquad J^{(j)} = \int_{W_j} F^{(j)}, \qquad J^{(0)} = 1. \tag{7}$$

In both cases $a - M/2 + q - j = -\kappa_j$ (direct substitution of a and q, for $N \ge H$ and for $N < H$).

Lemma 2 (proved)**.** *(a) If $j < q$ and $\kappa_j > 0$, then $Z^{(j)} < \infty$. (b) If $j \ge 1$ and $\kappa_{j-1} < 0$, then $J^{(j)} < \infty$, and $\int_{W_j,\, z_j < \theta} F^{(j)} \le c_j\, \theta^{-\kappa_{j-1}}$ for $0 < \theta \le 1$.*

Proof. On W, $y_l - y_{l'} \le y_l$ for $l < l'$. (a) $g^{(j)} \le \prod_l y_l^{a+r-l}(1+y_l)^{-M/2}$. Each factor is integrable on $(0,\infty)$: at 0 because $a + r - l \ge a > -1$, at ∞ because $a + r - l - M/2 \le -\kappa_j - 1 < -1$. (b) $F^{(j)} \le \prod_i z_i^{-\kappa_j + j - i} e^{-z_i/2}$. The worst factor is $i = j$; since $\kappa_j = \kappa_{j-1} + 1$, its exponent $-\kappa_j = -\kappa_{j-1} - 1$ exceeds -1, and its integral over $(0,\theta)$ is $\theta^{-\kappa_{j-1}}/(-\kappa_{j-1})$. The factors $i < j$ have exponents larger by at least one. □

Lemma 3 (proved)**.** *Let $j = q$, or $j < q$ with $\kappa_j > 0$. Then $\int_{W_{q-j}} g^{(j)} h_S^{(j)}\, dy = Z^{(j)}\, \Phi_j(S)$ for every S. For $C = \delta u v^\top$, $\Phi_j(s) = \mathbb{E}\, e^{-s\tau_U \tau_V}$ with independent $\tau_U \sim \mathrm{Beta}(\frac{N-j}{2}, \frac{j}{2})$, $\tau_V \sim \mathrm{Beta}(\frac{M-H+j}{2}, \frac{H-j}{2})$, where $\mathrm{Beta}(c,0) = \delta_1$ and $\mathrm{Beta}(0,c) = \delta_0$, and*

$$\Phi_j(s) = {}_2F_2\big(\tfrac{N-j}{2}, \tfrac{M-H+j}{2}; \tfrac{N}{2}, \tfrac{M}{2}; -s\big). \tag{8}$$

Proof. Case $j = q$. $\mathrm{tr}(SOE^{(q)}O^\top) = \frac{n}{2}\|(I - P_U)C\|_F^2$ with U the first q columns of O. If $q = N$ this vanishes and $\Phi_N \equiv 1$; if $q = H < N$ the V-average in Eq. (3) is over V_0 and $I - P_V = I_M$.

Case $j < q$. Let $p' = N - j$, $r' = H - j$, so $\min(p', r') = q - j$ and $|p' - r'| = |N - H|$. From $\kappa_j > 0$ and $j < q \le N$, $M - r' = M - H + j \ge N - j \ge 1$. For $\alpha \in \mathbb{R}^{p' \times r'}$, $\beta \in \mathbb{R}^{r' \times M}$, $\Gamma \in \mathbb{R}^{p' \times M}$ consider $I(\Gamma) = \iint e^{-\frac{1}{2}\|\beta\|^2 - \frac{1}{2}\|\alpha\beta - \Gamma\|^2} d\alpha\, d\beta$. Integrating β first (columnwise Gaussian, as in Lemma 1) gives $(2\pi)^{r'M/2} \int \det(I + \alpha^\top\alpha)^{-M/2} e^{-\frac{1}{2}\mathrm{tr}\,\Gamma^\top(I+\alpha\alpha^\top)^{-1}\Gamma} d\alpha$. Integrating α first (rowwise least squares; β has rank $r' \le M$ almost everywhere) gives $(2\pi)^{p'r'/2} \int e^{-\frac{1}{2}\|\beta\|^2} \det(\beta\beta^\top)^{-p'/2} e^{-\frac{1}{2}\|\Gamma(I-P_\beta)\|^2} d\beta$, with P_β the projector on the row space of β. By (W) and Lemma 2(a), $I(0) = c\, Z^{(j)} < \infty$, so by Tonelli both forms are finite and equal for every Γ. Dividing by $I(0)$,

$$\mathbb{E}_\alpha\, e^{-\frac{1}{2}\mathrm{tr}\,\Gamma^\top(I+\alpha\alpha^\top)^{-1}\Gamma} = \mathbb{E}_\beta\, e^{-\frac{1}{2}\|\Gamma(I-P_\beta)\|^2}, \tag{9}$$

where α has density $\propto \det(I + \alpha^\top\alpha)^{-M/2}$ and β has density $\propto e^{-\|\beta\|^2/2} \det(\beta\beta^\top)^{-p'/2}$. Write $\beta = (\beta\beta^\top)^{1/2} Q$; for Gaussian β the row frame Q is Haar and independent of $\beta\beta^\top$, and the reweighting depends on $\beta\beta^\top$ only, so $I - P_\beta = I - P_V$ with $V = Q^\top$ Haar on $V_{r'}(\mathbb{R}^M)$.

Now take $\hat{O}$ Haar on $V_{p'}(\mathbb{R}^N)$, independent, and $\Gamma = \sqrt{n}\, \hat{O}^\top C$. The right side of Eq. (9) becomes $\mathbb{E}\exp[-\frac{n}{2}\|(I-P_U)C(I-P_V)\|^2] = \Phi_j(S)$, with U the Haar j-frame complementary to $\hat{O}$. On the left side, (W) gives $(I + \alpha\alpha^\top)^{-1} = Q_\alpha\, \mathrm{diag}((1+y_l)^{-1}, 1_{p'-q+j}) Q_\alpha^\top$ with Q_α Haar on $O(p')$, independent of $y \sim g^{(j)}/Z^{(j)}$. The frame $\hat{O}Q_\alpha$ is Haar and independent of y; completing it to $O \in O(N)$ gives $\frac{1}{2}\mathrm{tr}\,\Gamma^\top(\cdot)\Gamma = \mathrm{tr}(SOE^{(j)}(y)O^\top)$, so the left side equals $\int g^{(j)} h_S^{(j)}/Z^{(j)}$.

Rank one: $\frac{n}{2}\|(I - P_U)C(I - P_V)\|^2 = s\,|(I - P_U)u|^2 |(I - P_V)v|^2$, and $|(I - P_U)u|^2$ is the squared norm of $N - j$ coordinates of a uniform point on S^{N-1}, hence $\mathrm{Beta}(\frac{N-j}{2}, \frac{j}{2})$; likewise for V. Expanding $e^{-s\tau_U\tau_V}$ and using $\mathbb{E}\tau^m = (c)_m/(c+d)_m$ for $\tau \sim \mathrm{Beta}(c,d)$ gives Eq. (8); the series converges absolutely. □

Lemma 4 (proved). *Suppose $\kappa_i > 0$ for $j \le i < q$. For $Y > 0$ and all $S \succeq 0$,*

$$T^{(j)}(Y,S) := \frac{1}{Z^{(j)}} \int_{W_{q-j},\, y_1 > Y} g^{(j)} h_S^{(j)} \le \frac{Z^{(j+1)}}{\kappa_j Z^{(j)}}\, Y^{-\kappa_j}\, \Phi_{j+1}(S), \tag{10}$$

and there are constants A_j with

$$\Phi_{j+1}(S) \le A_j\, (1+\operatorname{tr} S)^{\kappa_j}\, \Phi_j(S) \qquad \text{for all } S \succeq 0. \tag{11}$$

Proof. Write $y = (y_1, y'')$ with $y'' \in W_{q-j-1}$; then $g^{(j)}(y) = y_1^a (1+y_1)^{-M/2} \prod_{l\ge 2}(y_1 - y_l)\, g^{(j+1)}(y'')$, and $E^{(j)}(y) = E^{(j+1)}(y'') + (1+y_1)^{-1} e_{j+1} e_{j+1}^\top$.

Eq. (10): by (M), $h_S^{(j)}(y) \le h_S^{(j+1)}(y'')$; also $\prod_{l\ge 2}(y_1 - y_l) \le y_1^{q-j-1}$. Dropping the constraint $y_1'' \le y_1$ and using $\int_Y^\infty y^{a+q-j-1}(1+y)^{-M/2} dy \le \int_Y^\infty y^{-\kappa_j - 1} dy = Y^{-\kappa_j}/\kappa_j$ gives Eq. (10) by Lemma 3 at $j+1$.

Eq. (11), by downward induction on j. Let $t = \operatorname{tr} S$, $Y = K(1+t)$ with $K \ge 1$ to be fixed, and restrict the integral defining $Z^{(j)}\Phi_j(S)$ to $\{y_1 \ge 2Y,\ y_1'' \le Y\}$. There, $\prod_{l\ge 2}(y_1 - y_l) \ge (y_1/2)^{q-j-1}$ and, by (M), $h_S^{(j)}(y) \ge e^{-t/(1+y_1)} h_S^{(j+1)}(y'') \ge e^{-1/2} h_S^{(j+1)}(y'')$. With $1 + y \le 2y$ for $y \ge 1$, $\int_{2Y}^\infty y^a (1+y)^{-M/2} (y/2)^{q-j-1} dy \ge b_j\, Y^{-\kappa_j}$, $b_j > 0$. Hence

$$Z^{(j)}\Phi_j(S) \ge e^{-1/2} b_j K^{-\kappa_j} (1+t)^{-\kappa_j} Z^{(j+1)} \big[\Phi_{j+1}(S) - T^{(j+1)}(Y,S)\big]. \tag{12}$$

If $j+1 = q$, $T^{(q)} \equiv 0$. Otherwise Eq. (10) at $j+1$ and the induction hypothesis Eq. (11) at $j+1$ give $T^{(j+1)}(Y,S) \le c\, A_{j+1} K^{-\kappa_{j+1}} \Phi_{j+1}(S)$, which is at most $\frac{1}{2}\Phi_{j+1}(S)$ for K large. This proves Eq. (11) at j. □

4 Main theorem: unique dominant stratum

Theorem 5 (proved). *Let λ_j have a unique minimiser $j_* =: k$, and let*

$$\gamma = \tfrac{1}{3} \min\{1,\ \kappa_k \ (\text{if } k < q),\ -\kappa_{k-1} \ (\text{if } k \ge 1)\} > 0. \tag{13}$$

There are C_0, c_0, n_0, depending on N, H, M only, such that

$$\log R_n(C) \le \log \Phi_k(S) + C_0\, n^{-\gamma} \qquad \text{for all } n \ge n_0 \text{ and all } C, \tag{14}$$

$$\big|\log R_n(C) - \log \Phi_k(S)\big| \le C_0\big(\|C\|_F^{2\gamma} + n^{-\gamma/2}\big) \qquad \text{for } n \ge n_0,\ \|C\|_F \le c_0. \tag{15}$$

In particular $\lim_{\delta_0 \to 0} \limsup_{n\to\infty} \sup_{\|C\|_F \le \delta_0} |\log R_n(C) - \log \Phi_k(S)| = 0$, which covers $s \to 0$, $s = O(1)$ and $s \to \infty$ with $s \le \delta_0^2 n/2$.

Proof. Write $r = q - k$ and $y = (y_{\mathrm{top}}, y_{\mathrm{bot}}) \in W_k \times W_r$ with $y_{\mathrm{top},k} \ge y_{\mathrm{bot},1}$. By Lemma 2, $J := J^{(k)} < \infty$ and $Z := Z^{(k)} < \infty$.

Factorisation. $G_n(y) = G^{\mathrm{top}}(y_{\mathrm{top}}) \prod_{i\le k<l}(y_i - y_l)\, g^{(k)}(y_{\mathrm{bot}}) e^{-|y_{\mathrm{bot}}|/(2n)}$. Substituting $y_{\mathrm{top}} = nz$,

$$\int G^{\mathrm{top}} \prod_{i\le k<l} (y_i - y_l)\, dy_{\mathrm{top}} = n^e \int_{W_k,\ z_k \ge y_{\mathrm{bot},1}/n} F^{(k)}(z)\, \rho_n(z, y_{\mathrm{bot}})\, dz, \tag{16}$$

with $e = k(1-\kappa_k) + k(k-1)/2$ and $\rho_n = \prod_{i\le k}(1 + 1/(nz_i))^{-M/2} \prod_{i\le k<l}(1 - y_l/(nz_i)) \in [0,1]$.

Sandwich. By (M) and the ordering $(1+y_1)^{-1} \le \cdots \le (1+y_k)^{-1}$,

$$e^{-\operatorname{tr} S/(1+y_k)}\, h_S^{(k)}(y_{\mathrm{bot}}) \le h_S(y) \le h_S^{(k)}(y_{\mathrm{bot}}). \tag{17}$$

Upper bound. By Eqs. (17), (16) with $\rho_n \le 1$, and Lemma 3, $\int G_n h_S \le n^e JZ\,\Phi_k(S)$. For the denominator restrict to $z_k \ge \theta$, $y_{\mathrm{bot},1} \le \varepsilon' n$ with $\varepsilon' \le \theta^2 \le 1$. There $\rho_n \ge (1+1/(n\theta))^{-kM/2}(1-\theta)^{kr}$ and $e^{-|y_{\mathrm{bot}}|/(2n)} \ge e^{-r\varepsilon'/2}$, so by Lemma 2(b) and Eq. (10) at $S=0$,

$$\int G_n \ge n^e JZ\,\big[1 - c_k\theta^{-\kappa_{k-1}}/J\big]\big[1 - c\,(\varepsilon' n)^{-\kappa_k}\big](1+1/(n\theta))^{-kM/2}(1-\theta)^{kr}e^{-r\varepsilon'/2}. \tag{18}$$

The choice $\theta = n^{-1/3}$, $\varepsilon' = n^{-2/3}$ makes the deviations from one $O(n^{\kappa_{k-1}/3})$, $O(n^{-\kappa_k/3})$, $O(n^{-2/3})$, $O(n^{-1/3})$ and $O(n^{-2/3})$, all $O(n^{-\gamma})$, which gives Eq. (14). (Factors referring to the top block are absent if $k=0$, and factors referring to the bottom block are absent if $k=q$.)

Lower bound. Let $\varepsilon = \max(\|C\|_F^2/2, n^{-1/2})$, so $\operatorname{tr} S \le \varepsilon n$ and $\varepsilon n \ge 1$, and set $\theta = \varepsilon^{1/3}$, $\varepsilon' = \varepsilon^{2/3}$. On $\{z_k \ge \theta,\ y_{\mathrm{bot},1} \le \varepsilon' n\}$, $\operatorname{tr} S/(1+y_k) \le \varepsilon/\theta = \varepsilon^{2/3}$, and the bounds on ρ_n and $e^{-|y_{\mathrm{bot}}|/(2n)}$ used in Eq. (18) hold. Therefore

$$\int G_n h_S \ge n^e\big(J - c_k\theta^{-\kappa_{k-1}}\big)\,e^{-\varepsilon^{2/3}}(1+1/(n\theta))^{-kM/2}(1-\theta)^{kr}e^{-r\varepsilon'/2}\int_{y_1\le\varepsilon' n} g^{(k)}h_S^{(k)}. \tag{19}$$

The last integral is $Z[\Phi_k(S) - T^{(k)}(\varepsilon' n, S)]$, and Eqs. (10)–(11) give

$$T^{(k)}(\varepsilon' n, S) \le c\,A_k\Big(\frac{1+\varepsilon n}{\varepsilon' n}\Big)^{\kappa_k}\Phi_k(S) \le c'\varepsilon^{\kappa_k/3}\,\Phi_k(S). \tag{20}$$

The relative errors are $\varepsilon^{2/3}$, $\theta^{-\kappa_{k-1}} = \varepsilon^{-\kappa_{k-1}/3}$, $1/(n\theta) \le n^{-5/6}$, $\theta = \varepsilon^{1/3}$, ε' and $\varepsilon^{\kappa_k/3}$, all $O(\varepsilon^\gamma + n^{-5/6})$. Combining with $\int G_n \le n^e JZ$ gives $\log R_n \ge \log\Phi_k(S) - C(\varepsilon^\gamma + n^{-5/6})$, and $\varepsilon^\gamma \le (\|C\|_F^2/2)^\gamma + n^{-\gamma/2}$. The constants c_0, n_0 ensure that every bracket exceeds $\frac12$, so that logarithms of brackets are bounded by twice their deviation from one. □

Remark 6. The error in Eq. (15) does not vanish at fixed $C \ne 0$. For $(N,H,M) = (2,1,1)$ ($k=q=1$, $\Phi_1(s) = {}_1F_1(\frac12;1;-s)$), $\lim_{n\to\infty}[\log R_n - \log\Phi_1(n\delta^2/2)] = -\delta$ for every fixed $\delta > 0$ (proved). In Eq. (5), $G_n(y) = (1+y)^{-1/2}e^{-y/(2n)}$ and $h_s(y) = e^{-s/(1+y)}\,{}_1F_1(\frac12;1;-sy/(1+y))$. With $y = nz$, $s/(1+y) \to \delta^2/(2z)$, the ratio ${}_1F_1(\frac12;1;-sy/(1+y))/{}_1F_1(\frac12;1;-s) \to 1$ and is bounded on $y \ge 1$ by monotonicity of ${}_1F_1(\frac12;1;-x) = e^{-x/2}I_0(x/2)$, and the region $y < 1$ contributes $O(e^{-s/2}n^{-1/2}/\Phi_1(s)) \to 0$. Dominated convergence gives the limit $\int_0^\infty z^{-1/2}e^{-z/2-\delta^2/(2z)}dz/\sqrt{2\pi} = e^{-\delta}$. The value $-\delta$ is reproduced to 10^{-5} at $n \ge 10^{12}$ for $\delta = 0.01$–1 (checked numerically, `b9_checks.py h1`). For $(N,H,M) = (3,1,2)$ and $(2,1,3)$ the supremum over $s \le \varepsilon n$ of the error equals 0.447, 0.0447, 0.00447 at $\varepsilon = 10^{-1}, 10^{-3}, 10^{-5}$, i.e. $\sqrt{2\varepsilon} = \delta$, independently of $n \in [10^{10}, 10^{18}]$ (checked numerically). For $(3,2,3)$, $k=1$, the error at $s/n = 10^{-2}, 10^{-4}, 10^{-8}$ is -0.18, -0.025, -5×10^{-4} at $n = 10^{12}$ (checked numerically with `rrr.py`, whose contour rule limits the check to $s \le 10^{10}$). On the fibre $\{ab = C\}$, $\frac12|a|^2 + \frac12 b^2 \ge \delta$, so $e^{-\delta}$ is the largest prior weight there; Φ_k uses the prior at the origin (heuristic interpretation).

5 Large-s behaviour and learning coefficients

Lemma 7 (proved)**.** *Let $\tau_1 \sim \mathrm{Beta}(c_1,d_1)$, $\tau_2 \sim \mathrm{Beta}(c_2,d_2)$ be independent with $c_i, d_i > 0$, $T = \tau_1\tau_2$, $\nu = \min(c_1,c_2)$ and $\epsilon = 1$ if $c_1 = c_2$, $\epsilon = 0$ otherwise. Then $P(T \le x) \asymp x^\nu(1+\log\frac1x)^\epsilon$ for $0 < x \le 1$, and $\mathbb{E}e^{-sT} \asymp s^{-\nu}(1+\log s)^\epsilon$ for $s \ge 1$. If $\tau_2 \equiv 1$ the same holds with $\nu = c_1$, $\epsilon = 0$.*

Proof. $P(\tau_1 \le y) \asymp \min(1,y)^{c_1}$, so $P(T \le x) \asymp P(\tau_2 \le x) + x^{c_1}\mathbb{E}[\tau_2^{-c_1}; \tau_2 > x]$, and $\mathbb{E}[\tau_2^{-c_1}; \tau_2 > x] \asymp \int_x^1 t^{c_2-c_1-1}dt$. For the Laplace transform, $\mathbb{E}e^{-sT} = \int_0^\infty e^{-v}P(T \le v/s)\,dv$; the lower bound is $e^{-1}P(T \le 1/s)$, and the upper bound follows from $P(T \le v/s) \le C(1+v)^{\nu+1}P(T \le 1/s)$. □

Corollary 8 (proved). *Let $m = 1$, $k = j_*$, $C = \delta uv^\top$. Define ν_k and ϵ_k by Lemma 7 applied to the nondegenerate factors among τ_U, τ_V of Lemma 3 ($\nu_k = 0$ if $k = N$). Then:*

1. *$Z_n(0) = c\, n^{-\lambda_k}(1 + O(n^{-\gamma}))$, so $\lambda(0) = \lambda_k$ with multiplicity one.*
2. *For fixed $\delta \le c_0$, $\log Z_n(C) = -(\lambda_k + \nu_k)\log n + \epsilon_k \log\log n + O(1)$, so $\lambda(1) = \lambda_k + \nu_k$ and $m(1) = 1 + \epsilon_k$.*
3. *$\log \Phi_k(s) = -[\lambda(1) - \lambda(0)]\log s + [m(1) - 1]\log\log s + O(1)$ as $s \to \infty$.*

Proof. (1) By Lemma 1, $Z_n(0) = c\, n^{-q(a+1)-q(q-1)/2} \int G_n$, and the proof of Theorem 5 gives $\int G_n = n^e JZ(1+O(n^{-\gamma}))$. With $q = k+r$ and $2a+1 = |N-H|$ the total exponent is $-\frac{1}{2}[kM + r(r+|N-H|)] = -\lambda_k$. (2) Eq. (15) bounds $\log R_n - \log \Phi_k(n\delta^2/2)$ uniformly in $n \ge n_0$; insert Lemma 7. This determines the learning coefficient and multiplicity of every rank-one C with $\|C\|_F \le c_0$; Ref. [1] states that they depend on the rank of C only (cited). (3) is Lemma 7 restated with (1) and (2). □

The exponents can be written without case distinctions. With $g_j := \lambda_j + \frac{1}{2}(N-j)$ one has $\lambda_j + \frac{1}{2}(M-H+j) = g_{j+1}$, so $\lambda(1) = \min(g_k, g_{k+1})$ over the nondegenerate channels, and g is convex with its minimum at k or $k+1$ (proved: $g_{k+2} - g_{k+1} = \kappa_k + \frac{1}{2} > 0$ and $g_k - g_{k-1} = \kappa_{k-1} - \frac{1}{2} < 0$). Corollary 8 gives $\lambda(0)$ and $\lambda(1)$ independently of Ref. [1]. For $N, H, M \le 20$ (8000 triples) the values agree with `dln.lam` in exact arithmetic, and $m(1) = 2$ holds exactly when $N + H + M + 1$ is odd and the strict triangle inequalities $M + 1 < N + H$, $N + 1 < M + H$, $H + 1 < M + N$ hold (checked numerically, `b9_checks.py algebra`; this includes the $m = 2$ triples, treated with Theorem 9).

For truths of rank $\rho \ge 2$, Theorem 5 combined with Watanabe's asymptotics $\log Z_n(C) = -\lambda(C)\log n + O(\log\log n)$ at fixed C (cited: Ref. [3]) and with $\lambda(C) = \lambda(\rho)$ from Ref. [1] gives $\log \Phi_{j_*}(tS_0) = -[\lambda(\rho) - \lambda(0)]\log t + O(\log\log t)$ as $t \to \infty$ for every S_0 of rank ρ (proved, given the cited results). This replaces the codimension count of `derivations.tex`, Eq. (17), which remains heuristic as a direct statement about Φ_j.

6 Two dominant strata

Theorem 9 (proved). *Let $\lambda_{j_0} = \lambda_{j_0+1}$ ($m = 2$) and $C = \delta uv^\top$, $s = n\delta^2/2$, $\Phi := \Phi_{j_0+1}$. By Eq. (8), Φ_{j_0} and Φ_{j_0+1} coincide as ${}_2F_2$ functions because the tie condition $M - H + j_0 + 1 = N - j_0$ exchanges the two parameter pairs. There is C_0 such that for $n \ge 3$ and $0 \le s \le n/2$, with $\ell = \log \frac{n}{1+s}$,*

$$R_n(s) = \Phi(s)\,\frac{\ell + \xi_1}{\log n + \xi_0}, \qquad |\xi_0|, |\xi_1| \le C_0. \tag{21}$$

Consequently $\log R_n(s) - \log \Phi(s) - \log(\ell/\log n) = O(1/\ell)$ for $\ell \ge 2C_0$, which tends to zero uniformly as $n \to \infty$ and $\delta \to 0$, since $\ell \ge \min\{\log(n/2), \log(1/\delta^2)\}$.

Proof. Let $k = j_0$ and $r'' = q - k - 1$. Then $\kappa_k = 0$, $\kappa_{k-1} = -1$ and $\kappa_{k+i} = i$, so $J := J^{(k)} < \infty$, $Z := Z^{(k+1)} < \infty$, Lemma 4 applies for $j \ge k+1$, and Lemma 2(b) gives $J - J_\theta \le c\,\theta$. Write $y = (y_{\rm top}, Y, y_{\rm bot}) \in W_k \times \mathbb{R}_+ \times W_{r''}$, $Y = y_{k+1}$. The Y-dependent factors of G_n outside the top integral are $Y^a(1+Y)^{-M/2}e^{-Y/(2n)}\prod_l (Y - y_l)$, and $a + r'' - M/2 = -\kappa_k - 1 = -1$. The top integral obeys Eq. (16) with $F^{(k)}$ and is at most $n^e J$.

Step 1 (bottom cost). Let $\mu(Y, s) = Z^{-1}\int g^{(k+1)}(y_{\rm bot}) h_s^{(k)}(Y, y_{\rm bot})\, dy_{\rm bot}$. We show $\mu(Y, s) \le \Phi(s)\min\{1, C_1((1+Y)/(1+s))^{1/2}\}$. The bound $\mu \le \Phi$ follows from (M) and Lemma 3. With $w = O^\top u$ uniform on S^{N-1}, $h_s^{(k)} = \mathbb{E}_w \exp\{-s[w_{k+1}^2/(1+Y) + \mathcal{C}\,\hat{Y}]\}$, where $\mathcal{C} = \sum_{i>k+1} w_i^2$ and

$\hat{Y} \in [0,1]$ depends on $y_{\rm bot}$ and on the direction of $(w_i)_{i>k+1}$ only. The triple $(\sum_{i\le k} w_i^2, w_{k+1}^2, \mathcal{C})$ is Dirichlet$(\frac{k}{2}, \frac{1}{2}, \frac{N-k-1}{2})$ and independent of that direction. By the proof of Lemma 3 at $j = k+1$, $\hat{Y}$ averaged over $y_{\rm bot}$ has the law of $\tau_V^{(k+1)} \sim \mathrm{Beta}(\frac{N-k}{2}, \frac{H-k-1}{2})$ (tie condition used), and $\Phi(s) = \mathbb{E}e^{-s\mathcal{C}\tau_V^{(k+1)}}$ with $\mathcal{C} \sim \mathrm{Beta}(\frac{N-k-1}{2}, \frac{k+1}{2})$. On $\{\mathcal{C} \le \frac{1}{2}\}$, given $\mathcal{C}$, $w_{k+1}^2 = (1-\mathcal{C})\beta$ with $\beta \sim \mathrm{Beta}(\frac{1}{2}, \frac{k}{2})$ independent, and $\mathbb{E}e^{-s\beta/(2(1+\hat{Y}))} \le c\min\{1, ((1+Y)/s)^{1/2}\}$. On $\{\mathcal{C} > \frac{1}{2}\}$ the integrand is at most $e^{-s\hat{Y}/2}$, whose average is $O((1+s)^{-(N-k)/2})$, while $\Phi(s) \ge \mathbb{E}e^{-s\mathcal{C}} \ge c(1+s)^{-(N-k-1)/2}$ (Lemma 7). The ratio is $O((1+s)^{-1/2})$. Degenerate factors ($k = 0$, $k+1 = H$, or $k+1 = N$) only strengthen these bounds.

Step 2 (upper bounds). By (M), $h_s \le h_s^{(k)}(Y, y_{\rm bot})$. Bounding the top integral by $n^e J$, $\prod_l (Y - y_l) \le Y^{r''}$, $e^{-|y_{\rm bot}|/(2n)} \le 1$ and dropping $y_{{\rm bot},1} \le Y$,

$$\int G_n h_s \le n^e J Z\, \Phi(s) \int_0^\infty Y^{a+r''}(1+Y)^{-M/2} e^{-Y/(2n)} \min\Big\{1, C_1\Big(\frac{1+Y}{1+s}\Big)^{1/2}\Big\} dY. \tag{22}$$

Since $Y^{a+r''}(1+Y)^{-M/2} \le Y^{-1}$ for $Y \ge 1$ and $a \ge -\frac{1}{2}$, the integral over $Y \le 1$ is at most 2, over $1 \le Y \le 1+s$ at most $2\sqrt{2}\,C_1$, and over $Y \ge 1+s$ at most $E_1(\frac{1+s}{2n}) \le \ell + \log 3$, with E_1 the exponential integral. The case $s = 0$ bounds $\int G_n$ by $n^e J Z(\log n + C)$.

Step 3 (lower bounds). Restrict to $Y \in [K(1+s), n/K]$, $y_k \ge (nY)^{1/2}$, $y_{{\rm bot},1} \le ((1+s)Y)^{1/2}$, with K a large constant. By (M), $h_s \ge e^{-s/(1+y_k) - s/(1+Y)} h_s^{(k+1)}(y_{\rm bot})$, and $s/(1+y_k) + s/(1+Y) \le 2((1+s)/Y)^{1/2}$. In the top integral, $z_k \ge (Y/n)^{1/2}$ gives $\rho_n \ge 1 - c(Y/n)^{1/2} - cY^{-1/2}$ and $J_{(Y/n)^{1/2}} \ge J(1 - c(Y/n)^{1/2})$. The Y-factors are at least $Y^{-1}(1-c/Y)(1-Y/n)(1-((1+s)/Y)^{1/2})^{r''}$, and $e^{-|y_{\rm bot}|/(2n)} \ge 1 - r''Y/n$. The bottom integral is $Z[\Phi(s) - T^{(k+1)}(((1+s)Y)^{1/2}, s)] \ge Z\Phi(s)[1 - c((1+s)/Y)^{1/2}]$ by Eqs. (10)–(11) with $\kappa_{k+1} = 1$. All relative errors are $O(((1+s)/Y)^{1/2} + (Y/n)^{1/2})$, whose integral against dY/Y over $[K(1+s), n/K]$ is bounded. Hence $\int G_n h_s \ge n^e J Z\, \Phi(s)(\ell - C)$, and at $s = 0$, $\int G_n \ge n^e J Z(\log n - C)$. Steps 2 and 3 give Eq. (21). □

At $s = n^\theta$, Eq. (21) gives $\log R_n - \log \Phi \to \log(1-\theta)$. For $N = M = 3$, $H = 1$, $s = n^{1/2}$ the difference $D = \log R_n - \log \Phi$ is $-0.7171, -0.7049, -0.6990, -0.6960, -0.6946$ at $n = 10^8, 10^{16}, 10^{32}, 10^{64}, 10^{128}$, against $\log \frac{1}{2} = -0.6931$, with $D - \log(\ell/\log n)$ halving as $\log n$ doubles (checked numerically, `b9_checks.py h1`). At $s = 2$ the same difference is $-0.0313, \dots, -0.0019$, i.e. $O(1/\log n)$. For $(N, H, M) = (2, 2, 3)$, $D - \log(\ell/\log n)$ is $+0.023, +0.013, -0.021$ at $n = 10^{12}$, $s = n^{1/4}, n^{1/2}, n^{3/4}$, and $+0.014, +0.012$ at $n = 10^{20}$, $s = n^{1/4}, n^{1/2}$ (checked numerically with `rrr.py`).

7 What is not proved

1. Multi-scale form (heuristic). Theorem 5 holds for every rank of C, so for $m = 1$ the multi-scale statement of the outline (separate factors $\Psi_i(n\delta_i^2)$ for $\delta_1 \gg \delta_2 \gg \dots$) is equivalent to the factorisation of the n-independent function $\Phi_{j_*}(S)$ for widely separated eigenvalues of S. That factorisation is not proved here. A route is to apply Lemma 1 and the sandwich Eq. (17) to Φ_{j_*} itself after conditioning on the frame that captures the largest eigendirection; we have not carried it out.

2. Two dominant strata with rank ≥ 2 (not done). Theorem 9 uses rank one in Step 1, through the Dirichlet structure of a single vector w. For rank ≥ 2 the scale that replaces $1 + s$ in ℓ is not determined; with separated eigenvalues it may lie between the smallest and largest eigenvalue.

3. Explicit form of the fixed-δ correction for general (N, H, M) (not done). Remark 6 computes it for $(2, 1, 1)$ only.

4. Non-Gaussian priors (not done). Lemma 1 uses the $O(N)$-invariance of the prior of A and the Gaussian form of both priors.

References

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[3] S. Watanabe, *Algebraic Geometry and Statistical Learning Theory* (Cambridge University Press, Cambridge, 2009).